# 30054 International and Monetary Economics

## Spring 2020

## Exam (May 20, 2020)

**Do NOT open this exam until instructed to do so.**

**Problem 1 (Multiple choice questions)**

1. Gross national expenditure (GNE) is

   (a) GDP plus net factor income from abroad (NIFA)

   (b) GDP plus net unilateral transfers (NUT)

   (c) GDP minus trade balance (NX)

   (d) GDP minus consumption (C)

2. A Japanese resident purchases $100 million US Treasury bonds from a US resident. After this transaction, Japanese financial account balance defined as in our class (i.e., so that FA+KA+CA=0)

   (a) increases;

   (b) declines;

   (c) does not change;

   (d) None of the above.

3. The UIP states that domestic nominal interest rate equals foreign nominal interest rate

   (a) plus the expected rate of depreciation of domestic currency;

   (b) minus the expected rate of depreciation of domestic currency;

   (c) plus the expected rate of appreciation of domestic currency;

   (d) plus the expected rate of depreciation of foreign currency;

4. The AA schedule describes equilibrium on ...

   (a) candy market;

   (b) goods market;

   (c) foreign exchange market and money market;

   (d) None of the above.

5. When (non-US) domestic producers set their export prices in US dollars when they export their goods to countries other than the US, we call this

   (a) producer-currency pricing (PCP);

   (b) local-currency pricing (LCP);

   (c) dominant currency paradigm (DCP);

   (d) gross domestic product (GDP).

6. The absolute PPP theory of the nominal exchange rate determination implies that the real exchange rate equals

   (a) one;

(b) two;

(c) three;

(d) four.

7. Doom loop that we covered in class

   (a) proposes that self-fulfilling expectations can ruin both the financial and public sectors of a country;

   (b) is the name of a release of the video game Doom;

   (c) is statistical software that can be used to estimate regressions;

   (d) is the name of the stimulus program that the ECB launched to fight the effects of the coronavirus.

8. The theory of Optimum Currency Areas proposes that it makes more sense to establish a currency union when

   (a) countries labor markets are integrated;

   (b) countries goods markets are integrated;

   (c) countries experience symmetric shocks;

   (d) All of the above.

9. Which of the following groups of models that we discussed in class emphasize the importance of current or future fundamental economic conditions as opposed to the multiple equilibria situation in explaining balance-of-payments crises?

   (a) The 1st and 2nd generation models;

   (b) The 2nd and 3rd generation models;

   (c) The 1st and 3rd generation models;

   (d) None of the above.

10. Under The Bretton Woods System, all but one participating countries formally fixed their exchange rates to

   (a) The Italian Lira;

   (b) The British Pound;

   (c) The Soviet rubble;

   (d) None of the above.

**Problem 2 (TRUE/FALSE)**

Indicate if a statement is true or false. **Provide a very short justification.**

1. A simultaneously fixed exchange rate and an independent monetary policy are impossible.
2. Floating exchange rate regimes are superior to fixed exchange rate regimes.
3. Eurozone is an optimum currency area.
4. Expectations about the future do not matter for the exchange rate determination.
5. If the capital account of a country is zero ($KA = 0$), then a country can be a net capital receiver ($FA > 0$) and run positive current account surplus ($CA > 0$).

**Problem 3 (Analytic questions)**

1. Which of the following formulas represent the DD schedule discussed in class? (The notation is the same that we used in class)

   (a) $\overline{M}/\overline{P} = L[\overline{R}^* + (E^e - E)/E, Y]$ or, equivalently, $Y = Y(E, E^e, \overline{R}^*, P, \overline{M})$;

   (b) $Y = C(Y - \overline{T}) + \overline{I} + \overline{G} + NX\left(E\overline{P}^*/\overline{P}, Y - \overline{T}\right)$;

   (c) $R = \overline{R}^* + (E^e - E)/E$;

   (d) $FA + CA + KA = 0$.

2. Under the assumptions used in class (consumption responds less than one-to-one to disposable income and net exports increase in the real exchange rate), how does the DD schedule look like in the exchange rate/output space? Choose one correct answer from Figure 2.

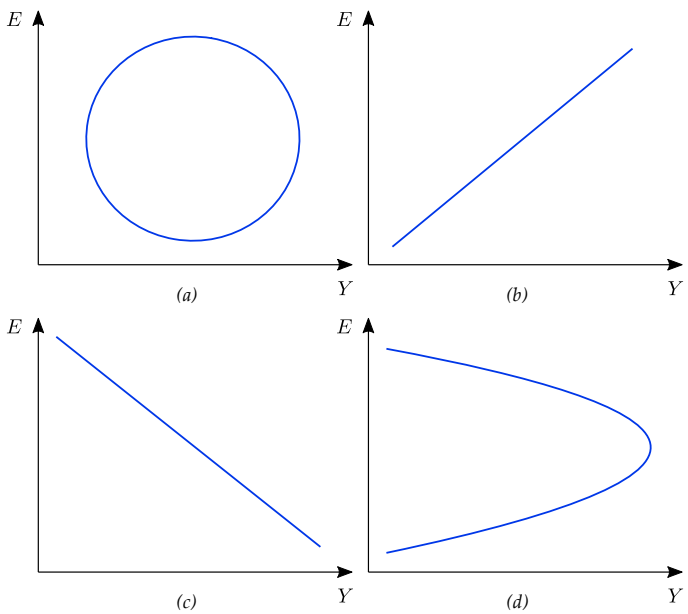


Figure 1: The DD schedule.

3. Which of the following formulas represent the AA schedule discussed in class? (The notation is the same that we used in class)

   (a) $\overline{M}/\overline{P} = L[\overline{R}^* + (\overline{E}^e - E)/E, Y]$ or, equivalently, $Y = Y(E, E^e, \overline{R}^*, P, \overline{M})$;

   (b) $Y = C(Y - \overline{T}) + \overline{I} + \overline{G} + NX\left(E\overline{P}^*/\overline{P}, Y - \overline{T}\right)$;

   (c) $R = \overline{R}^* + (E^e - E)/E$;

   (d) $FA + CA + KA = 0$.

4. Figure 2 represents the AA curve in the exchange rate/output space. Explain briefly why there is a negative relationship between output and exchange rate.

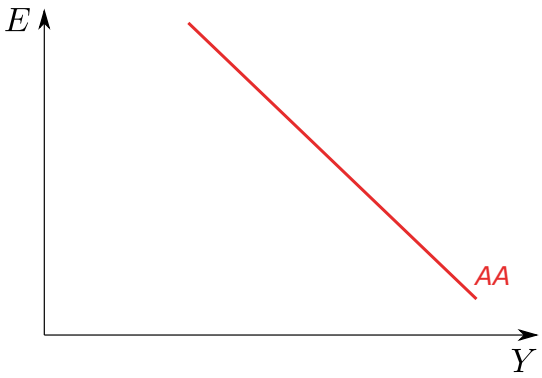


Figure 2: The AA schedule.

5. Which of the scenarios depicted in Figure 3 represents a response of the AA curve to a temporary expansion of the supply of money?

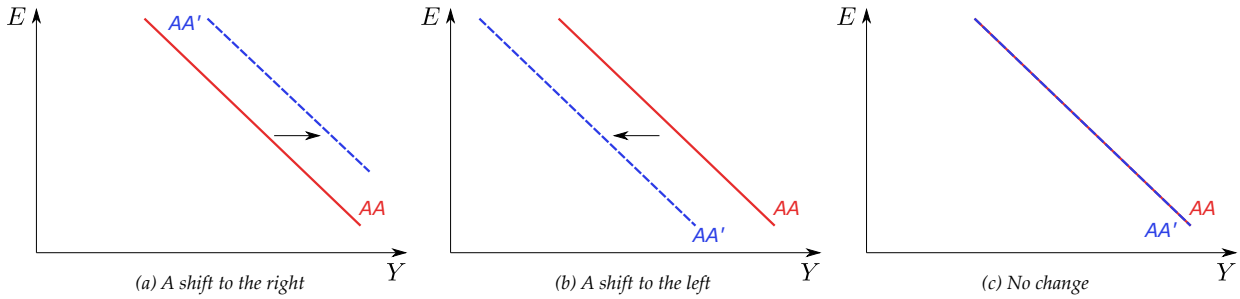


Figure 3: The AA schedule change after a temporary monetary policy expansion.

6. Figure 4 shows the AA schedule in the normal and the liquidity trap periods. Explain briefly why the AA curve becomes flat in the liquidity trap.

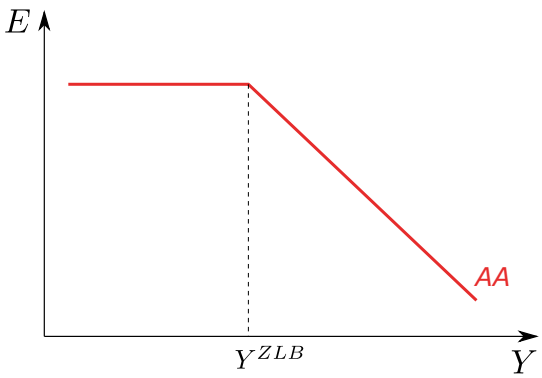


Figure 4: The AA schedule under liquidity trap and normal conditions.

7. Which of the scenarios depicted in Figure 5 represents a response of the AA schedule, which takes into account the liquidity trap, to a temporary expansion of the supply of money?

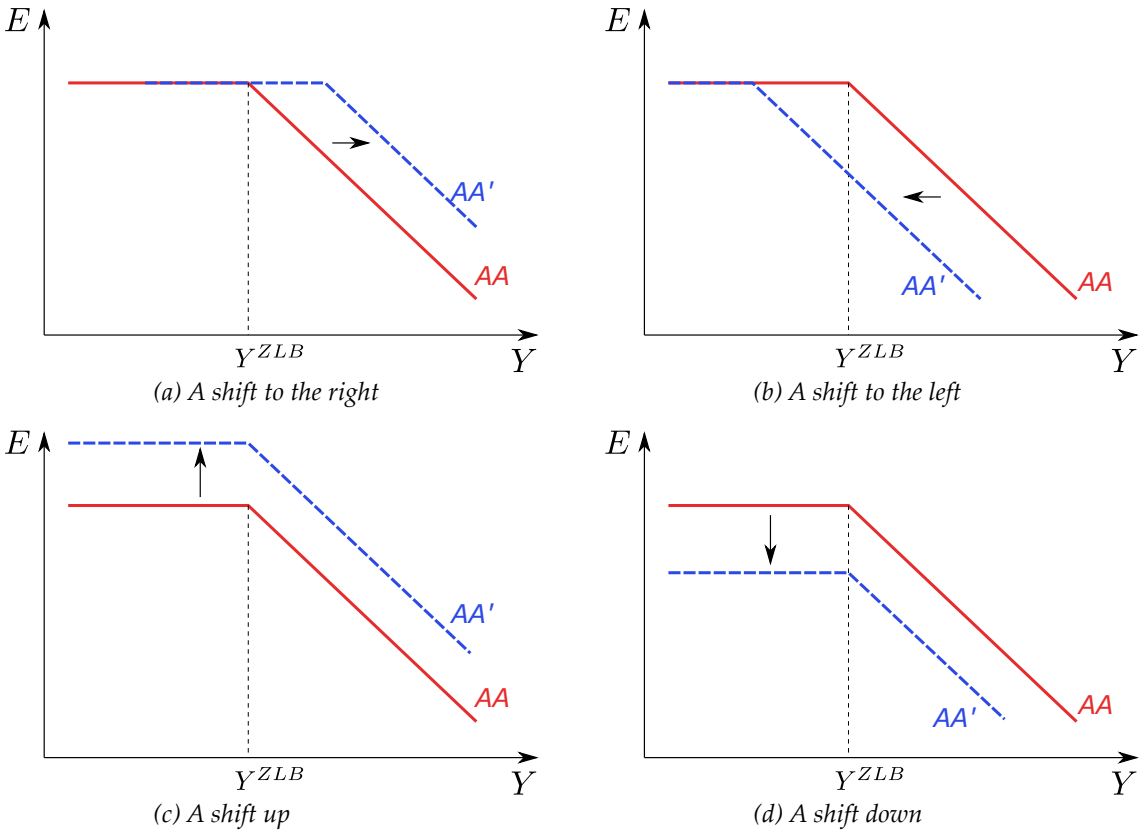


Figure 5: The AA schedule change after a temporary monetary policy expansion.

8. Use your answer to the previous question 7 and the assumption that the DD curve intersects the AA curve in its flat part, as depicted in Figure 6. A (small) temporary expansion of money supply

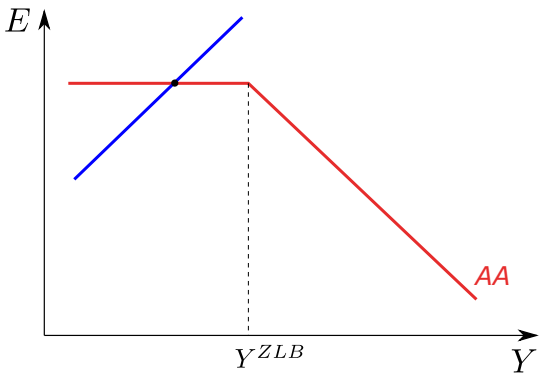


Figure 6: The AA and DD schedules in liquidity trap.

   (a) increases output;
   (b) decreases output;
   (c) does not change output;
   (d) all of the above.

9. Which of the scenarios depicted in Figure 7 represents a response of the AA schedule, which takes into account the liquidity trap, to central bank's promise

to increase the supply of money *permanently starting from the next year* (a form of forward guidance policy)?

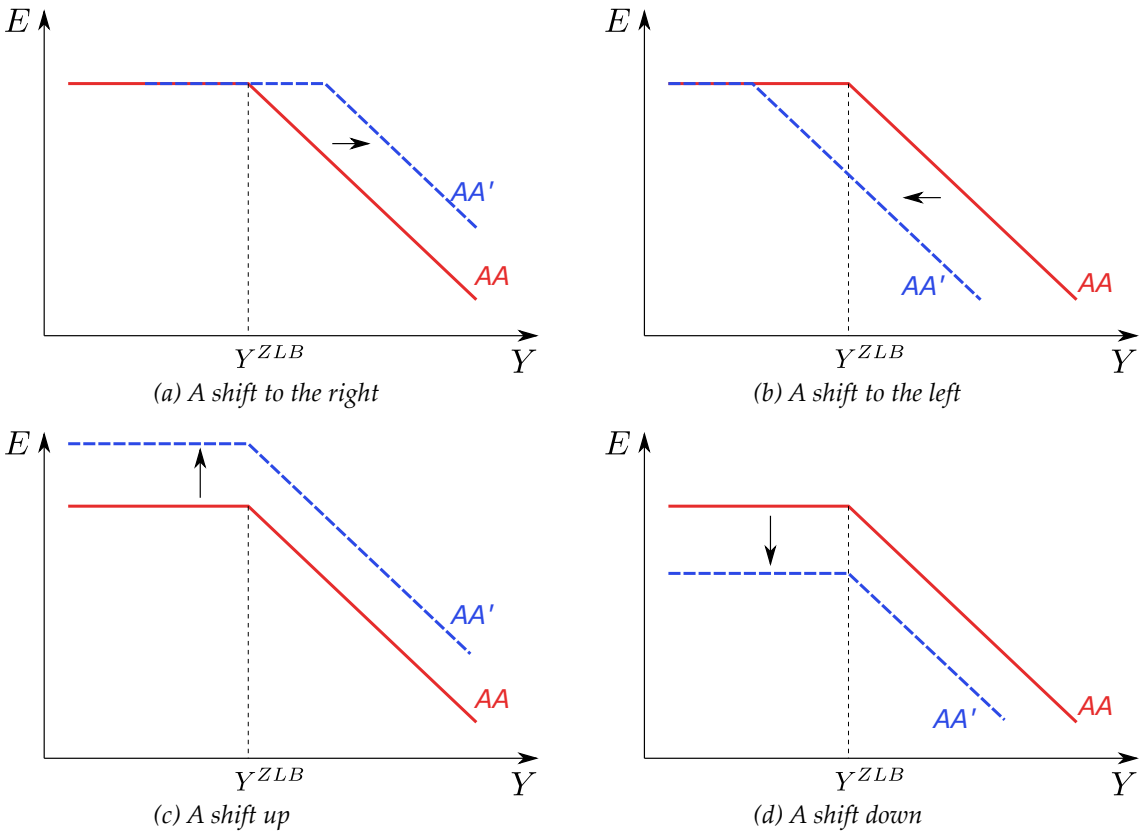


Figure 7: The AA schedule change after a promise of a future permanent monetary policy expansion.

10. Use your answer to the previous question 9 and the assumption that the DD curve intersects the AA curve in its flat part, as depicted in Figure 8. A central bank promise to permanently increase the supply of money in the next year by a small amount

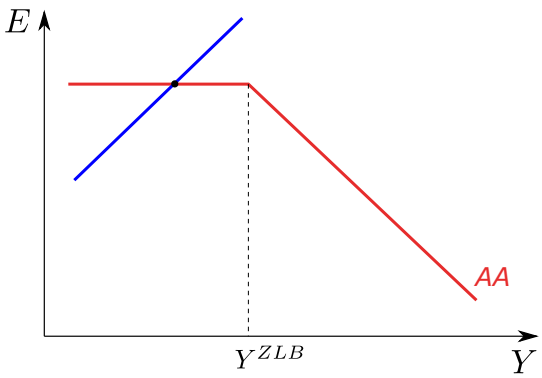


Figure 8: The AA and DD schedules in liquidity trap.

(a) increases output;

(b) decreases output;

(c) does not change output;

(d) all of the above.